

Lloyds Banking Group

Data Science & Analytics

Customer Retention Enhancement

Task 1: Data Gathering and Exploratory Analysis

SmartBank (subsidiary of Lloyds Banking Group)

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Executive Summary

This report covers Phase 1 of the SmartBank churn project. I have gathered the five supplied data sources, aggregated them to one row per customer, run exploratory analysis with appropriate statistical tests, and produced a cleaned dataset of 1,000 customers with 26 predictor features ready for Phase 2 modelling. The customer base shows a 20.4% churn rate.

No single feature in the supplied dataset is a strong predictor of churn. Login frequency has the strongest individual numerical association (point-biserial $r = -0.082$, $p = 0.010$), though the effect explains under 1% of churn variation. No categorical variable reaches the 0.05 significance threshold. A regularised logistic regression baseline scores ROC-AUC 0.514, barely better than chance. Three structural risks affect what Phase 2 can credibly conclude: a temporal-leakage concern around the churn measurement date, missing tenure and occupation fields, and a small positive sample of 204 churners.

1. Data Sources and Approach

All five sheets from the supplied workbook are in use. Customer_Service covers 668 of 1,000 customers, with the remaining 332 customers given zero service interactions (the absence itself reflects no service contact in the observation period). All other sheets cover all 1,000 customers, with no missing values or duplicates anywhere in the dataset.

Source	Rows	Unique customers	Role
Customer_Demographics	1,000	1,000	Age, gender, marital status, income level.
Transaction_History	5,054	1,000	Spend volume, value, recency, product mix.
Customer_Service	1,002	668	Complaint and resolution signals.
Online_Activity	1,000	1,000	Digital channel and login behaviour.
Churn_Status	1,000	1,000	Binary target label.

Table 1. Source sheets and their role in the analysis.

The analysis applies Wilson 95% confidence intervals for category churn rates, chi-square tests of independence (no Yates continuity correction, applied uniformly so figures and text match), Cramér's V for effect size, point-biserial correlations with Fisher-z 95% confidence intervals for numerical features, Mann-Whitney U for the bounded spend-share distributions, and variance inflation factors for multicollinearity.

2. Key Findings

2.1 Target variable

204 of 1,000 customers churned (20.4%). The 4 to 1 ratio rules out accuracy as a useful headline metric and shapes the choice of evaluation approach for Phase 2.

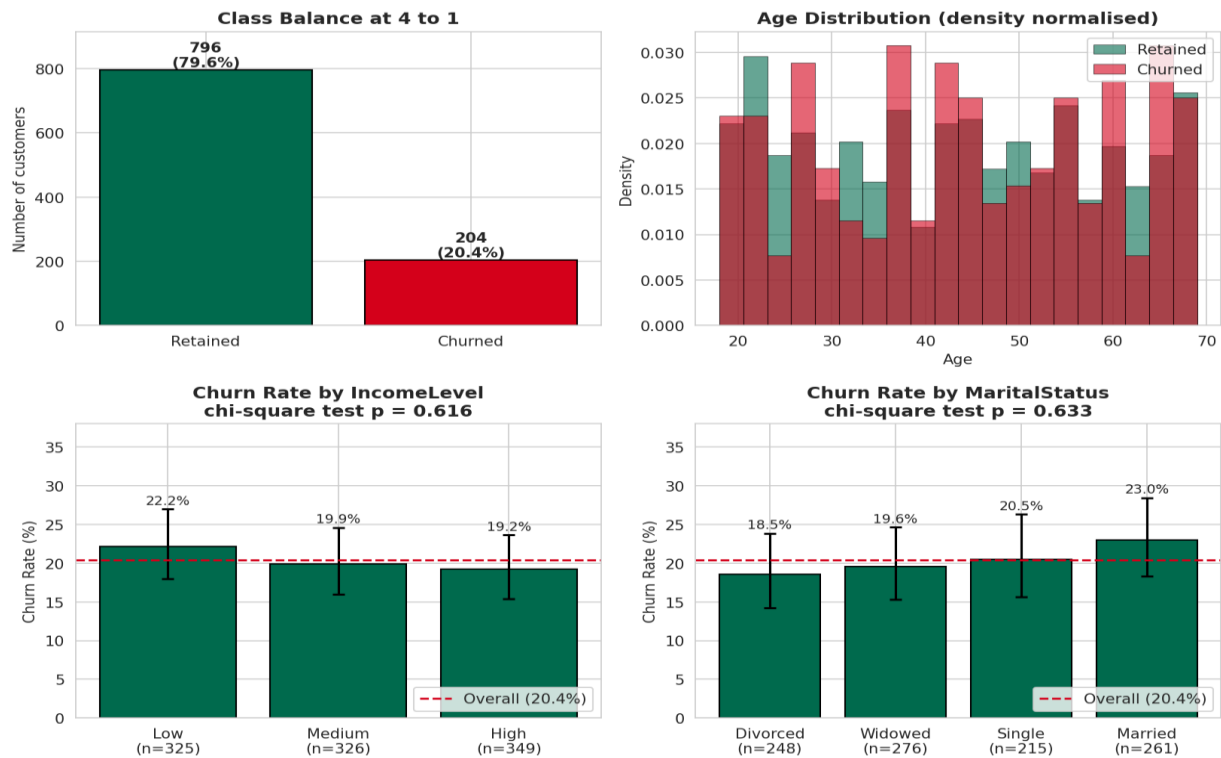


Figure 1. Class balance and demographic breakdowns. Error bars are Wilson 95% confidence intervals; chi-square p -values are shown above the bottom-row charts. None of the demographic differences crosses the 0.05 significance threshold.

2.2 Demographics show no significant associations

Group churn rates differ slightly. Low-income customers churn at 22.2% against 19.2% for high income, and married customers churn at 23.0% against 18.5% for divorced. The 95% confidence intervals overlap substantially in every case, with chi-square p -values of 0.616, 0.633 and 0.621 for income, marital status and gender. Cramér's V values are all under 0.05, meaning the effect sizes are negligible.

2.3 Behavioural patterns

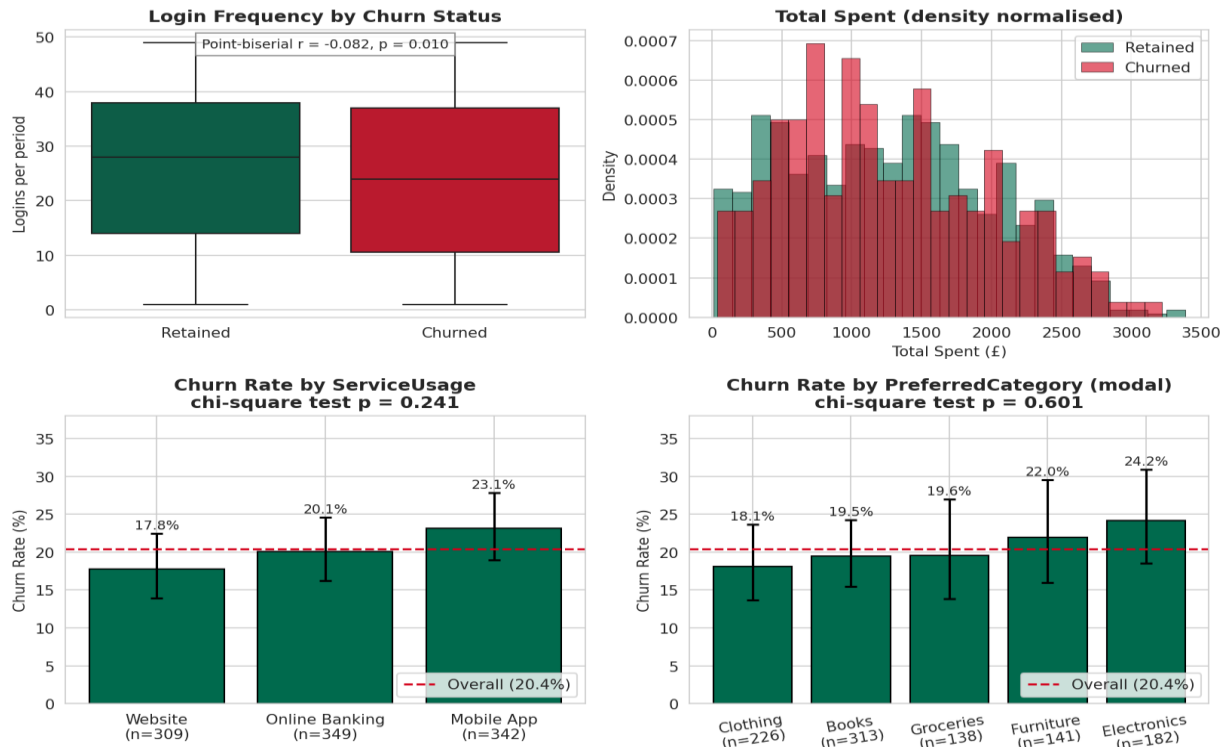


Figure 2. Behavioural patterns. Service channel and preferred category show visible differences, though the chi-square p -values exceed 0.24.

Login frequency has the strongest individual numerical association with churn (point-biserial $r = -0.082$, 95% CI -0.143 to -0.020, $p = 0.010$). The direction is intuitive since churners log in less often, with the small effect size tempering any expectation of a strong single-feature predictor. Service channel and preferred category produce chi-square p -values of 0.241 and 0.601 respectively, with effect sizes under 0.06 Cramér's V .

2.4 Spend-share patterns

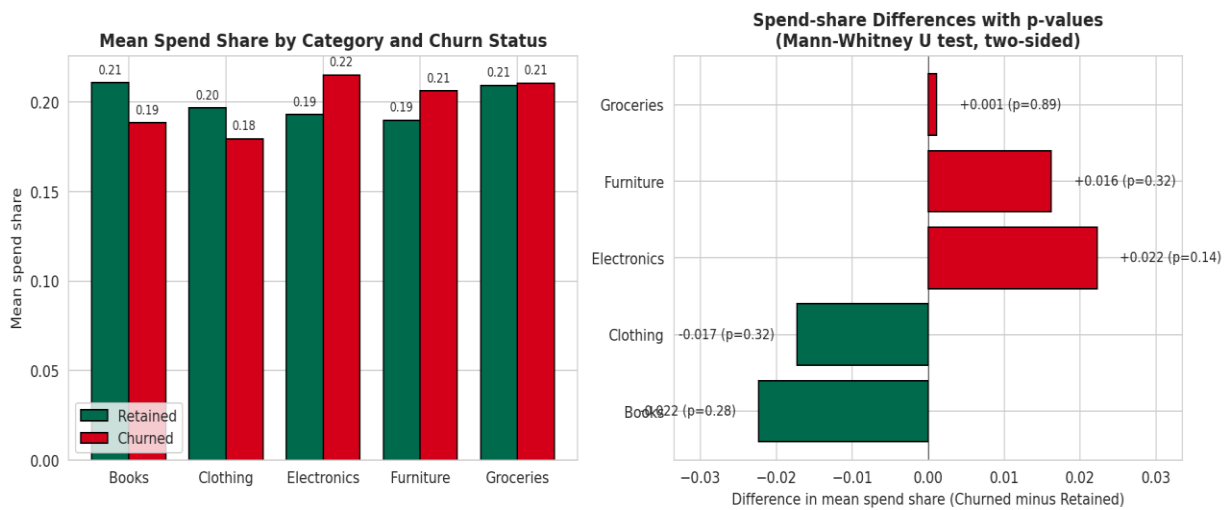


Figure 3. Mean spend share by category and churn status, with the difference and Mann-Whitney U p -value per category. Differences are small in every category and no p -value is below 0.14.

Churned customers spend a slightly higher share on Electronics and Furniture; retained customers spend a slightly higher share on Books and Clothing. Differences sit under three percentage points across every category. The shares are kept in the final dataset so a multivariate model can pick up any interactions with other features.

2.5 Service-friction signals

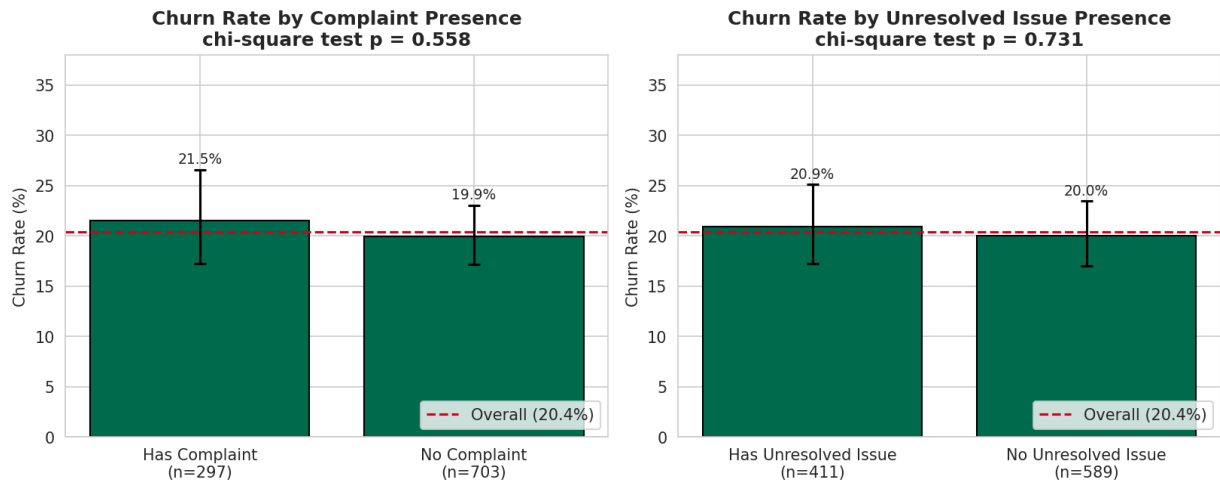


Figure 4. Churn rates for customers with versus without logged complaints and unresolved issues.

Customers with at least one complaint churn at 22.1% against 20.0% for those without. The chi-square p-value is 0.558 for complaints and 0.731 for unresolved issues. The dataset caps service interactions at two per customer, which limits the discriminative power any service-related feature can offer.

2.6 Severe multicollinearity in transaction features

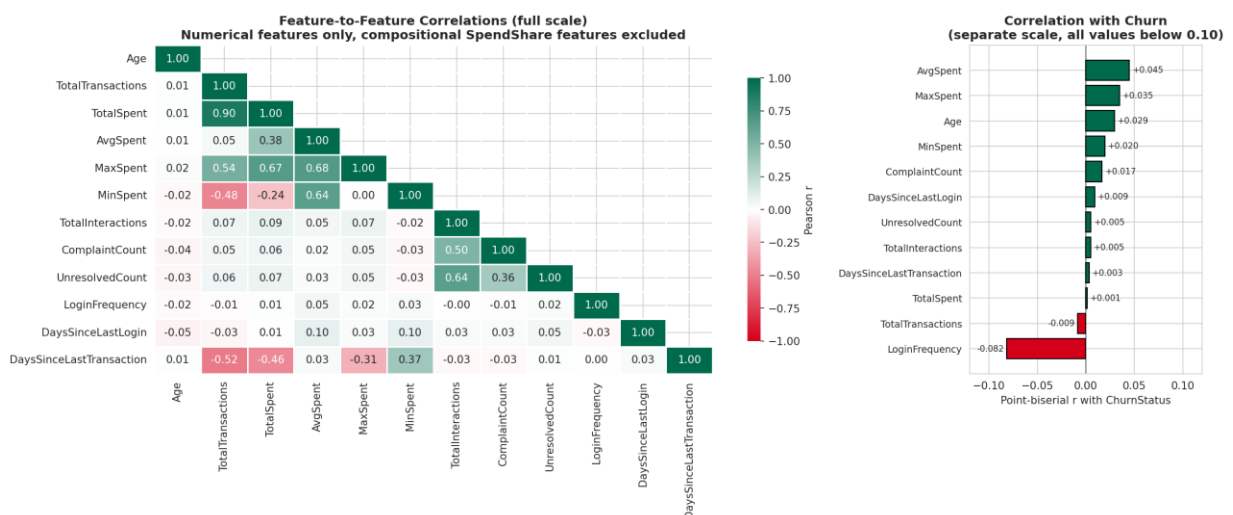


Figure 5. Left: full-scale feature-to-feature correlations. Right: separate-scale point-biserial correlations with churn. The two panels use different scales because feature-to-feature correlations reach 0.90 while target correlations stay below 0.10.

Feature	VIF	Interpretation
TotalSpent	18.4	Severe, drop or combine for linear models

Feature	VIF	Interpretation
AvgSpent	17.0	Severe, drop or combine for linear models
TotalTransactions	16.9	Severe, drop or combine for linear models
MaxSpent	6.8	Moderate, caution for linear models
MinSpent	5.9	Moderate, caution for linear models
Other features	< 2.0	Acceptable

Table 2. Variance inflation factors. The five SpendShare features are excluded from this table because they are compositional (sum to 1.0 per customer), so one of them must be dropped before any non-regularised linear model regardless of what the VIF would read.

TotalSpent is approximately TotalTransactions multiplied by AvgSpent by construction, so all three carry the same information. Linear models need L1/L2 regularisation or manual feature dropping. Tree models will fit without issue, though feature-importance rankings will be split arbitrarily across the redundant set, which is why the Modelling_Notes sheet asks for SHAP values rather than raw importances in Phase 2.

3. Limitations

Highest-priority risk: temporal leakage

Transaction dates run through 2022. Online-activity dates run through 2023. The churn label has no associated measurement date. If a customer churned during 2022, low 2023 login activity is a consequence of leaving rather than a predictor of it. Before Phase 2, the team must verify when ChurnStatus was measured and define an observation window for features that ends strictly before that date. The reference date used here (1 January 2024) is provisional.

Three further caveats affect what Phase 2 can credibly claim. The brief identifies working professionals and small-business owners as the target segments, while the supplied data has no occupation, employment status, business-account flag or salary-payment marker. Customer tenure is often an important churn predictor in retail banking and is also absent. With only 204 churners, every model metric will have a wide confidence interval. The data itself has several hallmarks of a synthetic or training sample (no missing values across 8,000+ records, uniform-looking distributions, transaction counts capped at nine), so findings should be treated as methodological practice rather than actionable market intelligence for SmartBank without validation on a real dataset.

4. Data Cleaning and Feature Engineering

The final dataset has 26 predictor features. Of these, 15 are engineered customer-level features: 11 from transactions (TotalTransactions, TotalSpent, AvgSpent, MaxSpent, MinSpent, DaysSinceLastTransaction, plus the five SpendShare_<Category> compositional features), 3 from customer service (TotalInteractions, ComplaintCount, UnresolvedCount), and 1 from online activity

(DaysSinceLastLogin). The remaining 11 features come from encoded demographic and channel variables (Age, Gender_M, IncomeLevel_Ord, four Marital_* dummies, three Channel_* dummies and LoginFrequency).

Encoding decisions follow the type of each variable: ordinal for income (Low=0, Medium=1, High=2), binary for gender, one-hot for marital status and channel. The PreferredCategory variable has been replaced by the five SpendShare features, which preserve the customer's spending mix rather than collapsing it to a single label.

Scaling is deferred to Phase 2

I have not fitted a scaler on this dataset. Fitting StandardScaler on all 1,000 rows before splitting into train and test sets causes train-test contamination, since the test fold's mean and standard deviation enter the training preprocessing. This form of data leakage is distinct from the target-leakage risk in section 3 (which concerns the churn measurement date). The deliverable workbook ships only the unscaled data. Scaling must be fitted on the training fold inside a scikit-learn Pipeline. The Modelling_Notes sheet of the workbook gives the exact pattern.

5. Recommendations for Phase 2

- Verify when ChurnStatus was measured and trim any feature whose measurement window extends past that point. This is the single most important step.
- Request occupation, employment status, SME-account flags, tenure and account-opening dates from the source system.
- Start with regularised baselines (logistic regression with L1 or L2, shallow decision tree) before non-linear models. The Task 1 baseline (ROC-AUC 0.514, PR-AUC 0.241) is a starting reference.
- Use proportional sampling on ChurnStatus for the train/test split. Apply SMOTE or other resampling inside cross-validation folds rather than before splitting.
- Report PR-AUC, precision, recall and F1 for the positive class. Report recall at top 10% if the retention team can only contact the riskiest decile. Include calibration through a reliability diagram and Brier score.
- Use SHAP values for the chosen model rather than raw importance rankings, given the multicollinearity in transaction features.
- Report performance separately for gender, income band and marital-status groups before deployment. Final feature use should be reviewed with legal, compliance and model-risk stakeholders.

Deliverables

- Cleaned_Customer_Churn_Data.xlsx with three sheets: Cleaned_Data (1,000 by 28), Data_Dictionary (19 feature entries with notes on multicollinearity and leakage risk), and Modelling_Notes (13 numbered guidelines for Phase 2).

- Task1_EDA_Report.docx, this report.
- Task1_EDA_Notebook.ipynb, the analysis notebook, which reproduces every figure, table and statistical test in the report end-to-end.